LAUREN LIAO

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EDUCATION

CARNEGIE MELLON UNIVERSITY, TEPPER SCHOOL OF BUSINESS — Pittsburgh, PA/USA
*Master of Science in Computational Finance – **Data Science Track*** ***GPA: 3.96/4.0*** — Expected 12/26

- Best Risk Management Award in RITC x CMU 2025 Trading Competition - volatility & algo trading cases

UNIVERSITY OF WATERLOO, MATH & BBA DOUBLE DEGREE PROGRAM — Waterloo, ON/Canada
*Bachelor of Mathematics – **Data Science** and **Mathematical Optimization** Double Major* ***GPA: 4.0/4.0*** — 4/22

- Participated in Poker club and Investment club activities

Bachelor of *BBA at Wilfrid Laurier University – **Finance** Specialization* **GPA: *3.9/4.0***

EXPERIENCE

TD SECURITIES — Toronto, ON
Model Developer, Quantitative Engineering & Development (5/24 – 7/25) — 5/21 – 8/21; 9/22 – 7/25

- Implemented valuation engines for treasury locks and total return swaps in C++; updated library based on business needs
- Developed and supported x-gamma risk tool for trading desks, involving python/vba programming and UI development
- Created Flask and Streamlit app using asynchronous programming in python to centralize and automate daily tasks
- Proposed and built CI/CD pipelines using GitHub Actions/Jenkins to streamline deployment of pricing models and unit tests

Global Markets Data Science Rotation Associate (9/22 – 4/24)

- *Equities Algo MM* - Researched on Hedge Fund activities and developed reactive strategies; Utilized Random Forests model to predict position imbalance; Researched on EOD market impacts for CRB using historical data, conducted factor analysis
- *Structured Notes Trading* - Created Dash app to visually analyze shocks & spreads, resulting in 60% in pricing efficiency
- *XVA* - Improved capital management efficiency by creating top of house view of counterparty risks and RAROC sheet

Summer Associate – Data Analytics (5/21 – 8/21)

- Backtested and analyzed ETF automated trading strategies utilizing Onetick, Excel, SQL and Python
- Generated weekly ETF Strategy reports of market focus and fund flows, streamlined processes to python, saving 50% time

PURPOSE INVESTMENTS — Toronto, ON
Quantitative and Data Analyst Intern — 1/21 – 4/21

- Collaborated with PMs & sales to develop visual market analytics on firm website using Pandas, Plotly, REST API and Git
- Researched on covered call strategies and reverse mortgage by backtesting in Python and conducting quantitative analysis

COMPETITIONS & PROJECTS

CMU ML - Bankruptcy Prediction with Fuzzy Shadowed SVM — 01/26-02/26

- Developed a robust bankruptcy prediction model for highly imbalanced datasets where bankrupt firms represented <5%
- Achieved a 19.3% increase in AUC-ROC and a 73.8% boost in G-Mean compared to traditional SVM baselines

CMU SCS - Algorithmic Trading Engine: Multi-Asset Exchange Simulator with Arbitrage *Bots* — 10/25 – 11/25

- Developed a multi-currency limit-order-book exchange simulator with FIFO matching, partial fills, priority-queue bid/ask
- Deployed an Arbitrage Bot using Bellman-Ford negative-cycle detection on log-price graphs, with optimal execution flow
- Designed full portfolio and P&L mechanics (inventory updates, fill handling, USD valuation) and a comprehensive test suite

Top 15 Finalist - National Undergraduate Big Data Challenge — 5/20 -7/20

- Applied Machine Learning algorithms (tensorflow, sklearn) and forecasting models (non-seasonal ARIMA, cross correlation) in Python and R to identify key factors for COVID-19 outbreak and predict future trends

ADDITIONAL INFORMATION

- Certificates: **Chartered Financial Analyst (CFA)**
- Programming skills: Python/VBA/SQL (proficient - numpy, pandas, sklearn, asynchronous programming), C++ (intermediate)
- Interests: Running half-marathons, Squash, Reading, Playing pipa, swimming